$\omega := 10000 \frac{rad}{s}$ $R_1 := 100\ \Omega$ $L_1 := 0.01\ H$ $R_2 := 50\ \Omega$

$C_1 := 0.00001\ F$

Del 1 - Impedanserne

Navne: R alene er gaskonstanten, V er volt og A er ampere i Mathcad, så alle størrelser har et subscript. Spolens og kondensatorens impedanser:

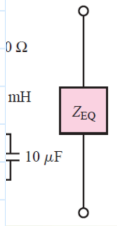
$\mathbf{Z}_L = j\omega L = \omega L \angle \frac{\pi}{2}$ Impedance of an inductor (3.56)

$Z_L := 1\mathrm{j} \cdot \omega \cdot L_1 = 100\mathrm{j}\ \Omega$

$\mathbf{Z}_C = \frac{1}{j\omega C} = \frac{-j}{\omega C} = \frac{1}{\omega C} \angle \frac{-\pi}{2}$ Impedance of a capacitor (3.61)

$Z_C := \frac{1}{(1\mathrm{j} \cdot \omega \cdot C_1)} = -10\mathrm{j}\ \Omega$

Impedance – Combined circuit



- **Find:** The equivalent impedance of the circuit Z_{EQ}
- Known quatities: ω = 10^4 rad/s; R1 = 100 Ω; L = 10 mH; R2 = 50 Ω; C = 10 μF
- The parallel impedance $Z_{||}$ of the R_2-C circuit:

$$Z_{||} = R_2 \left\| \frac{1}{j\omega C} = \frac{R_2(1/j\omega C)}{R_2 + 1/j\omega C} = \frac{R_2}{1 + j\omega C R_2} \right.$$

$$= \frac{50}{1 + j10^4 \times 10 \times 10^{-6} \times 50} = \frac{50}{1 + j5} = 1.92 - j9.62$$

$$= 9.81 \angle (-1.3734)\ \Omega$$

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Del 2 - Reduktion indefra og ud

Først R_2 parallelt med C_1 (produkt over sum, som for modstande).
ωCR_2 = 5 afgør resultatet:

$$\omega \cdot C_1 \cdot R_2 = 5 \qquad Z_p := \frac{R_2 \cdot Z_C}{\left(R_2 + Z_C\right)} = \left(1.923 - 9.615\text{j}\right)\ \Omega$$

Så serieleddene, lagt sammen i kartesisk form:

$$Z_{eq} := R_1 + Z_L + Z_p = \left(101.923 + 90.385\text{j}\right)\ \Omega \qquad \left|Z_{eq}\right| = 136.227\ \Omega \qquad \arg\left(Z_{eq}\right) = 41.566\ deg$$

Imaginærdelen (reaktansen) er positiv, så kredsløbet er induktivt ved
10 000 rad/s. Med 12 V i LTspice:

$$I_s := \frac{12\ V}{Z_{eq}} = \left(65.907 - 58.446\text{j}\right)\ mA \qquad \left|I_s\right| = 88.089\ mA \qquad \arg\left(I_s\right) = -41.566\ deg$$

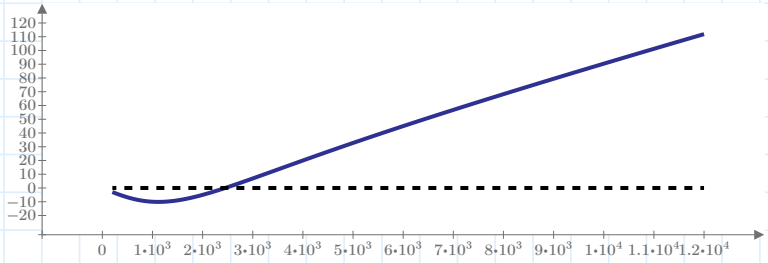
$Z_{eq} = \left(101.923 + 90.385\text{j}\right)\ \Omega$ $\left|Z_{eq}\right| = 136.227\ \Omega$ $\arg\left(Z_{eq}\right) = 41.566\ deg$

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Plot - Induktiv eller kapacitiv?

Svaret afhænger af frekvensen. Reaktansen som funktion af ω:

$$Z_\omega(w) := R_1 + 1\mathrm{j}\cdot w\cdot L_1 + \frac{R_2}{\left(1 + 1\mathrm{j}\cdot w\cdot C_1\cdot R_2\right)} \qquad w := 200\ \frac{rad}{s}, 250\ \frac{rad}{s}..12000\ \frac{rad}{s}$$



$w \left(\frac{rad}{s}\right)$

$\mathrm{Im}\left(Z_\omega(w)\right)\ (\Omega)$

$0\cdot\mathrm{Im}\left(Z_\omega(w)\right)\ (\Omega)$

Under ca. 2 450 rad/s er reaktansen negativ (kapacitiv), over er den positiv (induktiv). Ved 10 000 rad/s er den +90,4 Ω.

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